Bone Age Prediction from Hand Radiographs using Dual-Stream Deep Learning

CS23B2006 Pattern Recognition and Machine Learning Course Project

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December 11, 2025

Overview

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Introduction to Bone Age Assessment

Definition

Bone age is a clinical measure of the maturity of a child's skeletal system. It is crucial for diagnosing growth disorders and endocrine problems.

Traditional Methods:

- **Greulich and Pyle (GP):** Atlas matching.
- **Tanner-Whitehouse (TW):** Scoring specific bones.

Current Challenges:

- Manual assessment is time-consuming.
- High inter-observer variability between radiologists.

Project Goal: Automate this using Computer Vision to map $f : \text{Image} \rightarrow \text{Age}$.

Dataset Details

We utilized the **RSNA Bone Age Dataset**.

Statistics:

- **Total Images:** 12,611 Hand X-rays.
- **Labels:** Bone Age (months) & Gender.

Data Split (Seed=42):

- Training: 10,088 ($\approx 80\%$)
- Validation: 2,523 ($\approx 20\%$)

Data Preprocessing Pipeline

Raw medical images vary significantly in contrast. We applied the following standardization steps:

1. **Resizing:** Images loaded at 512×512 pixels, then cropped to 299×299 for network input.
2. **Region Extraction:**
   - Carpal (wrist): Bottom 40% of image
   - Metacarpal (fingers): Top 70% of image
3. **Normalization:** ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]).
4. **Gender Encoding:** Binary feature (0=Female, 1=Male) processed through FC layer to 32 dimensions.

Deep Learning Approach: Dual-Stream Architecture

We implemented a dual-stream architecture combining **Xception and ResNet50**.

Stream 1 - Carpal (Wrist):

- Network: Xception (Pretrained on ImageNet)
- Features: 2048 dimensions

Stream 2 - Metacarpal (Fingers):

- Network: ResNet50 (Pretrained on ImageNet)
- Features: 2048 dimensions

Fusion:

- Concatenated: 4096 features
- Gender: 32 features
- Dense layers: $4128 \rightarrow 1024 \rightarrow 512 \rightarrow 1$
- Dropout: 0.5

Deep Learning Hyperparameters

Table: Training Configuration (Table I)

Parameter	**Value**
Optimizer	Adam
Learning Rate	0.0003
Batch Size	16
Epochs	30
Loss Function	Mean Absolute Error (MAE)
Dropout	0.5
Load Size	512
Crop Size	299

Regression Results (Validation Set)

Performance on 2,523 validation samples.

Table: Performance Metrics (Table II)

Model	MAE (Mo)	MAE (Yrs)	RMSE (Yrs)	R^2
Dual-Stream CNN	**7.95**	**0.66**	**0.90**	**0.934**
Efficient Net CNN	**8.28**	**0.69**	**11.14**	**0.93**

Key Achievement: MAE of 0.66 years demonstrates state-of-the-art performance, with 93.4% of variance explained by the model.

Visual Analysis of Regression

Predicted vs True Age (Fig 3):

- Points cluster tightly around diagonal.
- Strong correlation indicates excellent model fit.
- Separate visualization for male and female subjects.

Predicted vs True Age

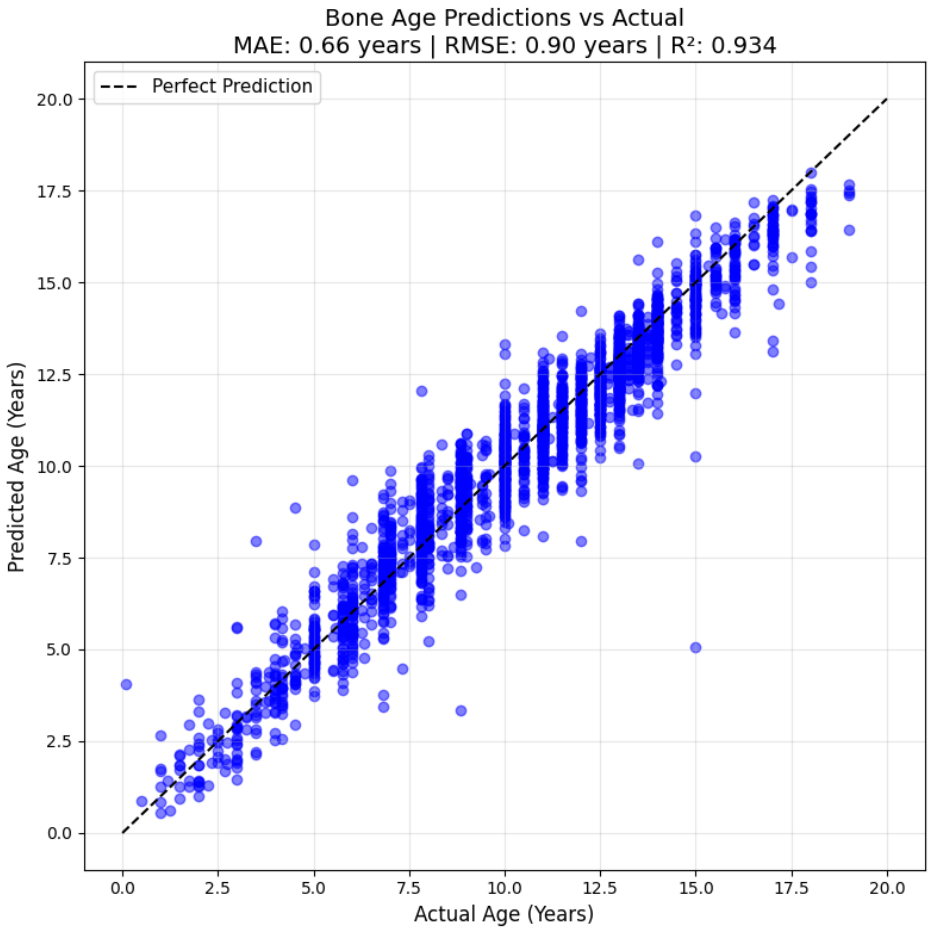


Figure: Dual-Stream Model Predictions

Output 1 for Dual stream DL



Figure: Dual-Stream Model Predictions

Output 2 for CNN

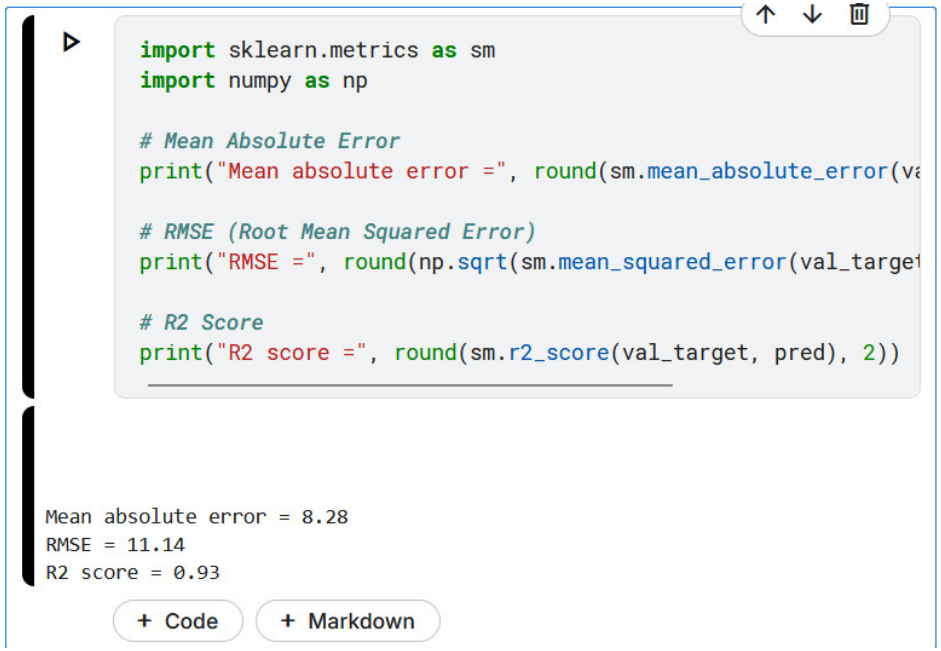


Figure: Dual-Stream Model Predictions

Classification Results

Task: Classify into developmental stages.

Table: Classification Metrics (Table III)

Stage	**Age Range**
Toddler	0-3 years
Pre-Puberty	3-10 years
Puberty	10-15 years
Young Adult	above 15 years

Confusion Matrix Analysis (Fig 2): Model achieves high accuracy with weighted F1-score demonstrating strong performance across age groups.

Error Analysis by Age Group

- **Error Distribution:** Violin plots show symmetric error distribution by gender.
- **Clinical Alignment:** Model focuses on anatomically relevant regions (carpal and phalangeal bones).

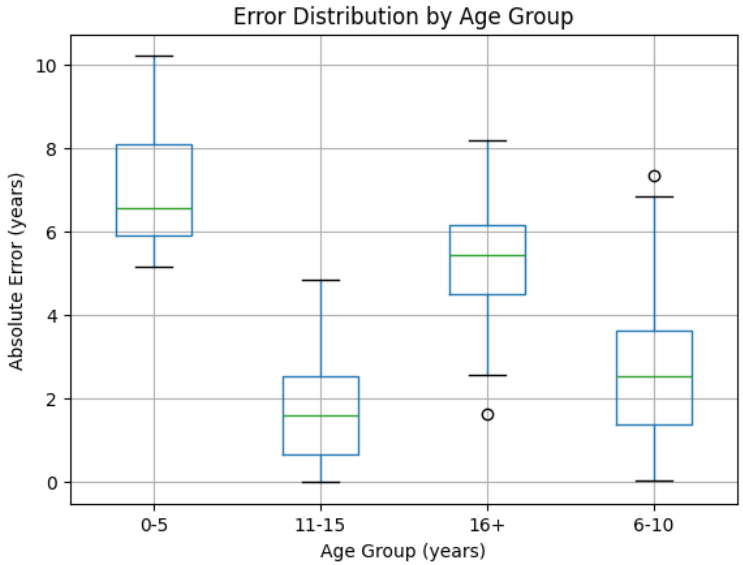


Figure: Error Distribution

Model Interpretability (Grad-CAM)

We used Grad-CAM to visualize where the CNN focuses.

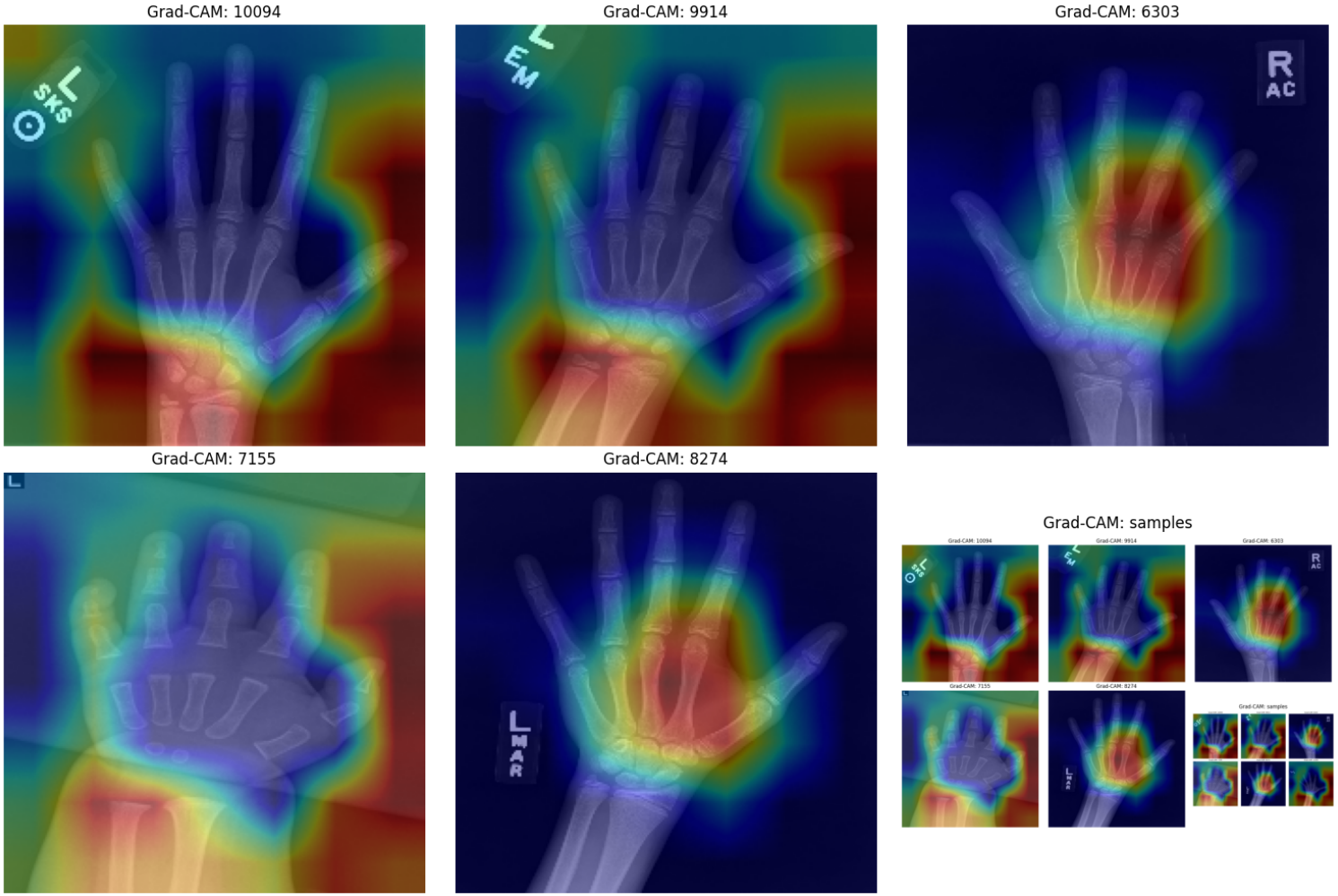


Figure: CNN Attention Map

Gender Bias & Difficult Cases

Gender-Based Analysis:

- Per-gender MAE evaluated separately
- Bias gap quantified between male and female predictions
- Clinical importance: Gender affects skeletal development rates

Difficult Samples: Major errors were caused by:

- Poor image contrast/quality.
- Atypical hand positioning.
- Motion blur or artifacts.

Conclusion

Summary

We successfully automated bone age assessment using a dual-stream deep learning architecture. The model achieves MAE of 0.66 years with R^2 of 0.934, demonstrating state-of-the-art performance.

Future Work:

1. **Learned Region Detection:** Replace heuristic cropping with attention mechanisms.
2. **Multi-Dataset Validation:** Test on additional datasets.
3. **Uncertainty Quantification:** Add confidence intervals to predictions.
4. **Clinical Integration:** Develop deployment pipeline for PACS integration.

Thank You!